



Diurnal versus nocturnal pollination in a subalpine wetland: From network structure to plant reproduction

Yue Teng^{a,b}, Soraya Villalobos^c, Jana C. Vamosi^d, Xiao-Fan Wang^{a,b,*}, Yan-Bing Gong^{a,b,*}

^a College of Life Sciences, State Key Laboratory of Hybrid Rice, Wuhan University, Wuhan 430072, China

^b Key Laboratory of Biodiversity and Environment on the Qinghai-Tibetan Plateau, Ministry of Education, Wuhan University, Wuhan 430072, China

^c Department of Basic Sciences, Universidad del Atlántico, Barranquilla 018000, Colombia

^d Department of Biological Sciences, University of Calgary, Calgary T2N1N4, Canada

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ABSTRACT

Diurnal and nocturnal pollinators are fundamental for the functioning of natural ecosystems and food security. However, differences between diurnal and nocturnal pollination networks and the contribution of these pollinators to plant fecundity have not been assessed comprehensively within a community context. In the Dajihu subalpine wetland of central China, we supplemented data from an intensive field survey with the analysis of pollen loads carried by diurnal and nocturnal pollinators to compare the structures of diurnal and nocturnal pollination networks using network parameters. Simultaneously, we compared the contribution of diurnal and nocturnal pollinators to plant reproduction of the nine dominant species based on stigma pollen placement. We found that although diurnal and nocturnal pollination networks exhibited a similar structural organisation, the diurnal network displayed greater stability and broader interaction with pollinators and plants compared to the nocturnal network. However, nocturnal pollinator species tended to establish less generalised interactions with the available floral resources. Both diurnal and nocturnal pollinators played a significant role in plant reproduction, and nocturnal pollination can result in greater pollen germination. In terms of stability, the lower redundancy of links within nocturnal networks suggests they may be more vulnerable to the temporary or permanent disappearance of plant resources in the study area. Our findings highlight the differences in the contributions of diurnal and nocturnal pollinators to plant reproduction at the community level, demonstrating that nocturnal pollinators are not redundant in maintaining species diversity in communities. This knowledge should play a pivotal role in preventing future pollination declines and improving pollinator conservation efforts.

1. Introduction

Pollinators play a key role in maintaining plant diversity by influencing sexual reproduction in terms of species numbers, genetic variation, and functional group richness (Wei et al., 2021), which ultimately contribute to ecosystem services associated with plant diversity. However, pollinators are declining worldwide, resulting in concerns over parallel declines in food production and plant

* Corresponding authors at: College of Life Sciences, State Key Laboratory of Hybrid Rice, Wuhan University, Wuhan 430072, China
E-mail addresses: wangxf@whu.edu.cn (X.-F. Wang), ybgong@whu.edu.cn (Y.-B. Gong).

diversity which depend on essential pollination services (Vamosi et al., 2006; Biesmeijer et al., 2006). The main factors causing this decline are land-use change, biotic invasions, pesticides, climate change, and the rapid increase in global artificial light at night (Potts et al., 2016; Knop et al., 2017; Briolat et al., 2021). Among these, artificial light at night affects the physiology and behaviour of a variety of organisms and has an impact on species communities and population dynamics (Gaston and Bennie, 2014). The research has also shown that because nocturnal and diurnal pollinators act as complementary functional groups and are linked by plant-mediated indirect interactions, artificial light at night has the potential to further aggravate the decline of diurnal pollinators through plant-mediated indirect effects from the nocturnal pollinator community, with further knock-on effects on plant pollination (Giavi et al., 2021; Knop et al., 2017). In addition, climate change has also been a topic of widespread concern in recent years, due to the enormous and increasing impacts on ecosystems caused by global warming (Byers, 2017; Cirtwill et al., 2022). Some researchers have argued that pollen viability is highly sensitive to heat stress, as temperatures above 30 °C may result in decreased viability (Vara Prasad, 1999; Mesihovic et al., 2016). As global temperatures increase, it is conceivable that nocturnal pollination during cooler periods may lead to a relatively elevated rate of pollen germination at night. This is likely due to the fact that pollen acquired through nocturnal activity of pollinators remain unexposed to high temperatures, potentially stabilising fruit and seed production (Corbet et al., 2020). Therefore, to develop appropriate management strategies for the conservation of pollination services, we need to better understand the structure of diurnal and nocturnal pollinator-plant interactions in the context of declining pollinators, and the respective contributions of pollinators to plant reproduction in communities.

Although researchers have begun to note the importance of nocturnal pollination networks, few studies have focused on diurnal and nocturnal pollinators simultaneously. Most studies have focused only on diurnal pollinators, especially on bees (Hymenoptera) and syrphids (Diptera) (Wood et al., 2015; Orford et al., 2015), ignoring numerous groups of nocturnal pollinators. However, studying nocturnal pollination networks may reveal a number of plant taxa, insect species, and unique interactions that differ from diurnal networks (Devoto et al., 2011). For instance, nocturnal settling macro-moths in agro-ecosystems have highly complex, formerly unknown plant interactions, with moths regularly transporting pollen of plants found in highly managed landscapes (Walton et al., 2020). These studies demonstrate that, at least in some communities, nocturnal pollination plays a more important role in sexual plant reproduction than previously suspected. Many studies have demonstrated that nocturnal pollination is at least equally important as, or highly complementary to diurnal pollination (Jennersten and Morse, 1991; Jaca et al., 2020), with nocturnal pollination contributing significantly to fruit production (Fijen et al., 2023). Nevertheless, previous investigations comparing the contribution of diurnal and nocturnal pollinators to plant reproduction have been limited to a single species. The contribution of diurnal and nocturnal pollinators to plant reproduction has yet to be determined at the community level.

The Dajiuhe wetland is a subalpine peat bog with high scientific value owing to its biodiversity, typicality, and representativeness (Luo et al., 2015). We chose the Dajiuhe wetland as a survey site to obtain information on plant-pollinator interactions in the region, to fill the research data gap, and to contribute to the conservation of species diversity in this area. In this study, we compared the components and structures of diurnal and nocturnal pollination networks using network parameters and compared the contribution of diurnal and nocturnal pollinators to plant reproduction based on stigma pollen placement. Specifically, we evaluated the following questions: 1) What are the differences in the components and structures between nocturnal and diurnal pollination networks? 2) To what extent do diurnal and nocturnal pollinators contribute to plant reproduction? 3) Is there variation in the contribution of diurnal and nocturnal pollinators among different plant species? We expect that there should be large taxonomic differences between diurnal and nocturnal pollinators due to the different capabilities of night vision among pollinator guilds. Likewise, the contribution of diurnal and nocturnal pollinators to plant reproduction may vary among species and could be determined by pollination networks.

2. Materials and methods

2.1. Study site

This study was conducted during the peak flowering season (from mid-July to late August) of 2018 in a subalpine wetland in Dajiuhe, Shennongjia National Park (31°30'15"N, 109°59'47"E, 1750 m a.s.l.) in western Hubei Province, central China. The site is situated at a relatively high altitude and features a climate marked by a short, warm, and wet summer and a long, cold winter. All the plants we studied were insect-pollinated, and most of them had overlapping floral phenologies.

2.2. Diurnal and nocturnal pollinator surveys

We observed plant-pollinator interactions during five intervals throughout the season. For the diurnal observations, each of these five intervals were comprised of 24 15-min observation periods for a total of 6 h per interval. Each census took place over 3–4 consecutive days and was separated from the start of the next census by 3–5 days. All surveys took place in six meadows that cover an excess of c. 5000 m² and are c. 200 m apart (Table A.1). Each meadow was divided into four quadrants. We randomly selected one of four quadrants within each meadow during each 15-min observation period, sampled the remaining quadrants in random order, and then repeated this in the other five meadows. We alternated the starting meadow on successive days. During each 15-min observation period, we walked around the focal quadrant at a steady speed and recorded all observed plant-pollinator interactions. We defined an interaction as taking place when a floral visitor of any species unambiguously contacted the reproductive structures of a flower. All observations took place between 0900 and 1600 h during weather conditions favourable for pollinator activity (i.e., low wind speed, not snowing or raining) (Table A.2). When a plant-pollinator interaction was observed, the timer was stopped and each flower visitor was caught individually using a hand net and immediately transferred to a separate vial to avoid pollen contamination from contacting

with other collected insects for subsequent pollen analysis and identification. For each collected visitor, the plant species was also recorded.

Nocturnal flower visitors were sampled using light traps baited with 160 W mercury tubes powered by 12 V batteries, and sampling nights were selected based on suitable weather conditions. Since most visitors at night were active from sunset to midnight, and the number of Lepidoptera was highest at midnight (Knop et al., 2018), the traps were strategically placed on the path closest to the meadows and operated from 2000 to 2400 h. Six light traps were used, each with an interval of 5–10 days (Table A.2). Captured insects were retained in individual tubes and stored in a freezer until pollen analysis (Bosch et al., 2009; Devoto et al., 2011) and identification.

The flower visitors were determined to the genus or family level, and, where possible, to the species level, using a combination of morphological and molecular techniques (DNA barcoding methods with the CO1 gene) (Table A.3). We divided all visitors into 10 functional groups: (1) bumblebees (*Bombus* spp.); (2) honeybees (*Apis cerana*); (3) solitary bees (Halictidae, Melittidae, *Eucera* spp., Andrenidae, Braconidae, Ciceridae, Scolidae, Ichneumonidae, Pemphredonidae, Tenthredinidae); (4) hoverflies; (5) other flies; (6) beetles; (7) stinkbugs and cicadas; (8) butterflies; (9) moths; and (10) others (ants, mosquitos, Orthoptera, Mecoptera, Trichoptera). All species of insects were prepared as pinned specimens and stored in the Gong laboratory at Wuhan University.

2.3. Pollen analysis

To facilitate pollen analysis, a pollen reference collection was prepared by sampling pollen from each flower species present at the sites, fixing them on microscope slides, and taking photos under a microscope (Binocular Nikon E100 microscope, Tokyo, Japan) at $\times 400$ magnification. To sample the pollen carried by flower visitors, insects were softened in a relaxing chamber for 24 h before handling (Devoto et al., 2011). We added 70% ethanol to submerged insect bodies in tubes (Hymenoptera were washed without corbicula and Lepidoptera were washed without wings). Pollen grains attached to the bodies of these insects were detached using an ultrasonic cleaner for 4 min (40 kHz; JP-010 T, Skymen Cleaning Equipment Shenzhen Co., Ltd., Shenzhen, China) (Gong et al., 2016). We washed each insect 1–10 times to ensure all pollen grains were removed. The pollen suspension was then placed into 2 mL tubes, which were then centrifuged for 2 min, until all the pollen was deposited on the bottom. The supernatant was discarded, and Alexander's solution was added to the centrifuge tube. The solution was mixed with a vortex oscillator for 30 s, removed with a pipette, dropped on a slide, and then examined under a microscope for identification and counting of pollen grains. We established the presence of at least five pollen grains from a given plant species in our pollen counts as proof of visitation to that species (Knop et al., 2017). Due to the limited sample sizes, a subset of pollinators (52 species) was prioritized for specimen preparation and identification, and consequently, pollen analysis was not conducted on these species. Among the insects subjected to pollen analysis, only four species had recorded pollen of less than five grains. However, as the sample size for these four insects species in pollen analysis was limited to ≤ 2 individuals, their potential roles as pollinators cannot be definitively ruled out. Only eight insect species were captured and deemed pollinators during both diurnal and nocturnal periods. Although we did not record the timing of pollen deposited on the insect body, we assumed that the diurnal and nocturnal activity of those insects meant that they were likely to deliver pollen to plants in both periods.

2.4. Plant abundance

Within each meadow quadrant, we randomly established nine $2 \times 2 \text{ m}^2$ plots to measure floral abundance (Table A.4). Simultaneous with pollinator surveys, all open flowers or floral visual units were counted for all species present in each plot once per round (Gong and Huang, 2009). We considered a floral visual unit, which acted as a functional unit for pollinator attraction, to be a solitary flower (e.g., *Geranium rosthornii*) or an inflorescence (e.g., *Inula hupehensis*) with individual florets clustered together and which cannot be easily counted in the field. The average abundance of each plant in the sample was calculated (excluding plant species for which no visitors were recorded). Each plant species was collected in triplicate and preserved as specimens in the Herbarium of Wuhan University.

2.5. Estimation of pollination service

To understand the effects of pollinators on plant reproduction during the day and night, we conducted pollination experiments on nine dominant plant species within the region in 2018. The high relative abundance of these plants in the community should prevent the influence of low abundance from affecting the results. The experiment was conducted a total of three times (with each round of experimentation commencing on July 20th, August 3rd, and August 15th) during the flowering season. Each round took place over 6–8 consecutive days and was conducted as follows: before the onset of flowering, more than 100 plants of each species were randomly selected, and their flower heads were bagged with a white nylon mesh (mesh size: 0.5 mm) to exclude flower visitors. This resulted in a total of 100 observed plants subjected to four treatments (approximately 25 plants in each treatment were distributed equally among the sites). The four treatments were: (1) plants were permanently bagged to assess the level of autogamy (hereafter referred to as the “Autogamy” treatment), (2) diurnally pollinated plants were bagged during the night (from 19:30 p.m. to 7:30 a.m. the next morning; hereafter referred to as the “Day” treatment), (3) nocturnally pollinated plants were bagged during the day (from 7:30 a.m. to 19:30 p.m.; hereafter referred to as the “Night” treatment), and (4) plants were left open to pollinators during 24 h of the day (hereafter referred to as the “Control” treatment). At the end of the flowering season, we opened the bags and collected the flowers. All flowers were stored in centrifuge tubes containing formalin, acetic acid, and alcohol (FAA, 5:5:90) (Gong and Huang, 2014). To facilitate the counting of

stigma pollen loads, we randomly chose 5–10 flowers of each species present at the three timepoints at six different sites. Each stigma was stained with Alexander's solution and squashed on a microscope slide to force the pollen into a single layer. Conspecific pollen grain deposition (CPD)/germination (CPG) was then identified and calculated based on size, shape, and a pollen reference collection of the nine studied species (Kearns and Inouye, 1993) (Table A.7).

2.6. Network analysis

The data were used to construct two networks. (i) The diurnal network (V_dP_d): was first constructed using the frequency of interactions between visitor groups and plant species based on all five rounds of observations (V_d), and second, using the diurnal pollen analysis (P_d). Because the diurnal pollen analysis could not include the pollen collected by rare pollinators and there were differences in the interactions recorded by the two sampling methods (Fig A.1), we attempted to fill these gaps with the visitation data we had, a procedure similar to that employed in Ballantyne et al. (2015). As the two networks cannot be combined directly owing to the difference in the two sampling methods, each visit frequency for a plant-visitor group interaction was calculated as a proportion of the total number of visits by all visitor groups, so that we obtained two networks of which the sum of the interaction bar widths was 1 for each plant species. Finally, we summed the two proportional networks and took the average to obtain an average network. The sum of all interaction bar widths for each plant species was 1. (ii) The nocturnal network (P_n) was constructed using nocturnal pollen analysis, with each plant-visitor interaction calculated as a proportion of the total nocturnal pollen data summed across all visitor groups for that plant, so that all interaction bar widths for each plant species sum to 1.

The following network-level parameters were calculated: plant species, animal species (Bersier et al., 2002), interactions recorded, connectance (i.e., the realised proportion of possible links), Shannon's diversity index (i.e., which measures interaction richness), interaction evenness (i.e., a measurement of homogeneity in interaction frequencies, reaching 1 when the number of interactions between plants and visitor groups is uniformly distributed), weighted NODF (i.e., a nestedness version that considers interaction frequencies, high values indicate nestedness) (Beckett et al., 2014), and H'_2 (i.e., a measure of specialisation, ranging between 0 (no specialisation) and 1 (complete specialisation)) (Dormann, 2011). To assess the significance of the two network metrics (weighted NODF and H'_2), we compared the observed values to those generated by null models. We controlled the effect of the network size by standardising the network links using Z scores against 1000 random network models ('r2dtable' function, Vegan package R (Dormann and Strauss, 2014)). The generalism of plants (i.e., the weighted mean effective number of species of flower visitors per plant species) and the generalism of visitors (i.e., the weighted mean effective number of plant species per pollinator species) was also calculated. We further selected two species-level parameters: species strength (i.e., the sum of dependencies of each species) and d' (i.e., species-level specialisation, which measures the exclusivity of interactions that individual species take part in). As the matrix data were proportional, all values were multiplied by 1000 before calculating those metrics (Ballantyne et al., 2015). Network-level and species-level parameters were calculated using the *networklevel* and *specieslevel* functions of the R package bipartite (Dormann et al., 2008). These techniques reduce the effects arising from differences in methodology in the diurnal pollen analysis data (P_d) and nocturnal pollen analysis data (P_n) and allow for a comprehensive comparison of network parameter calculations to understand the effects of time of day on potential network structure (Table A.5; Table A.6).

2.7. Statistical analysis

As differences in pollen production vary between species, it may affect the number of conspecific pollen grains deposited and germinated on stigmas. To simultaneously compare diurnal and nocturnal pollen deposition/germination on stigmas among different species, the number of pollen grains deposited/germinated of the nine dominant plant species was divided by the number of pollen grains per flower for standardisation. Since the pollen production of single flowers was represented as an average, we utilised the mean values of pollen deposition and germination on each plant stigma for subsequent comparisons. This value and the species-level parameters were taken as the response variable and analysed using linear mixed-effects (LME) models with the *lmer* function in the *lme4* package (Table A.8; Table A.9) (Bates et al., 2015). Treatment (two levels, diurnal versus nocturnal) was included as a fixed factor, and species (plants or animals) was included as a random factor. All LME models were visually checked for a normal distribution of residuals and homoscedasticity, and, if necessary, the dependent variable was transformed to satisfy model assumptions.

Generalised linear mixed models (GLMM) with a Poisson distribution were used to compare the pollen deposition and germination of stigmas among the different treatments for each of the nine plant species, followed by a Tukey test of multiple comparisons (Table A.10). The number of pollen grains deposited or germinating was analysed separately as a response variable, the different treatments (Autogamy, Day, Night, Control) were analysed as a fixed factor and each plant as a random factor.

We identified three 24-h plants (*Anaphalis aureopunctata*, *I. hupehensis*, and *Erigeron annuus*) where both nocturnal and diurnal pollinators play a significant role in their reproduction to calculate the pollen germination rate (the proportion of delivered pollen grains that were observed to germinate). Mixed-effects (LME) models were used to compare the pollen germination rate of stigmas between the day and night, with the pollen germination rate serving as the response variable, and the treatments of Day and Night being analysed as fixed factors. Each plant was included as a random factor. All analyses were performed using R version 4.1.1 (R Development Core Team, 2021).

3. Results

3.1. Comparing the structure of diurnal and nocturnal networks

3.1.1. Species composition

For the diurnal network, 115 pollinator and 41 plant species were recorded, representing 582 specific interactions. Diurnal pollen-based network (P_d) provided evidence for 413 interactions. Of these, 183 were common to the diurnal observation dataset (V_d) and 230 were new. In contrast, 169 interactions were only recorded during the field surveys (Fig. A.1). For the nocturnal network, 46 pollinator and 18 plant species were recorded, representing 96 specific interactions (Table 1). While diurnal networks are richer in interactions and require increased sampling, the rarefaction curves showed that our techniques to combine sampling resulted in relatively complete characterisation of interaction richness in diurnal and nocturnal networks (Fig. A.2). In the diurnal pollination network (Fig. 1a), Hymenoptera and Diptera were involved in more than 94% of the interactions. Bumblebees were involved in 13%, honeybees in 9%, solitary bees in 21%, hoverflies in 17%, and other flies in 35% of all interactions. Moths were the dominant group in the nocturnal network (46% of all interactions) (Fig. 1b). A total of 41 plant species from 19 families were recorded in both the diurnal and nocturnal pollinator networks (Table 1). Two groups of plant species were identified: the first group comprised 23 species, referred to as diurnal plants (visitation dropped to zero during the night), and the second group consisted of 18 species, referred to as 24-h plants (visitation did not drop to zero during the night).

3.1.2. Network-level parameters

The connectance and Shannon diversity index values (Table 1) indicate that the diurnal network contains greater levels of complexity than the nocturnal network in the study site. The interaction evenness varied little between the two networks. The weighted NODF for the diurnal network was approximately three times higher than for the nocturnal network (Table 1), although, given concerns regarding nestedness values for relatively small network sizes, this disparity should be treated with caution. The two networks were significantly more specialised than would be expected based on chance (Table 1); therefore, the nocturnal network appears as specialised as the diurnal network. Visitor and plant generality were separately 3.29 and 1.64 times higher in the diurnal network than in the nocturnal one (Table 1). Similar results also appeared in the comparison of network-level parameters between the P_d matrix and P_n matrix (Table A.5). It seems that the differences between the diurnal and nocturnal networks were not influenced by the sampling method or our standardisation of the networks.

3.1.3. Species-level parameters

There were no significant differences in species strength (the sum of dependencies of each species, which includes both the number and intensity of species interactions) between diurnal and nocturnal networks for plants ($t = -0.8$, $df = 15$, $P = 0.42$) or animals ($t = 1.5$, $df = 7$, $P = 0.19$) (Fig. A.3). The species strength (for both plants and pollinators) of the diurnal network showed large differences among species but there were fewer differences in the nocturnal network. There were significant differences in species-level specialisation (d') between diurnal and nocturnal networks for plants ($t = 7.7$, $df = 15$, $P < 0.001$) and pollinators ($t = 6.7$, $df = 157$, $P < 0.001$) (Fig. A.3). All pollinators and plant species in the nocturnal network showed relatively high specialisation. When comparing diurnal and nocturnal species-level parameters using only pollen-based networks (Fig. A.4; Table A.9), we observed a decrease in species strength for plants in the P_d network. This decline was likely attributed to the absence of certain rare pollinators in the diurnal pollen analysis data.

The species strengths of the nine plant species and their pollinators in the diurnal network were relatively higher than those in the nocturnal network, and the species-level specialisations were higher in the nocturnal network (Table 2). Of the nine plants in the diurnal network, three species (*G. rosthornii*, *Polygonum nepalense*, and *Sanguisorba officinalis*), for which no nocturnal visits were recorded, had a higher species strength of visitors during the day. Plant species had relatively low specialisation levels in the diurnal

Table 1

Network-level parameters describing the structure of the Dajihu pollination network based on diurnal and nocturnal plant-pollinator analysis. Weighted NODF is a nestedness version that considers interaction frequencies, high values indicate nestedness, and H'_2 is a measure of specialisation, ranging between 0 (no specialisation) and 1 (complete specialisation).

	Diurnal network	Nocturnal network
Plant species	41	18
Pollinator species	115	46
Interactions recorded	582	96
Connectance	0.124	0.116
Shannon diversity	5.384	4.178
Interaction evenness	0.637	0.622
Weighted NODF	27.315 (20%)** $P = 0.00$	8.113 (20%)** $P = 0.00$
H'_2	0.602 (20%)** $P = 0.00$	0.790 (20%)** $P = 0.00$
Generality of pollinators	8.405	2.558
Generality of plants	8.062	4.923

**Percentage of network descriptor value relative to the mean of the descriptor for 1000 randomly generated networks of the same size determined with r2dtable algorithm.

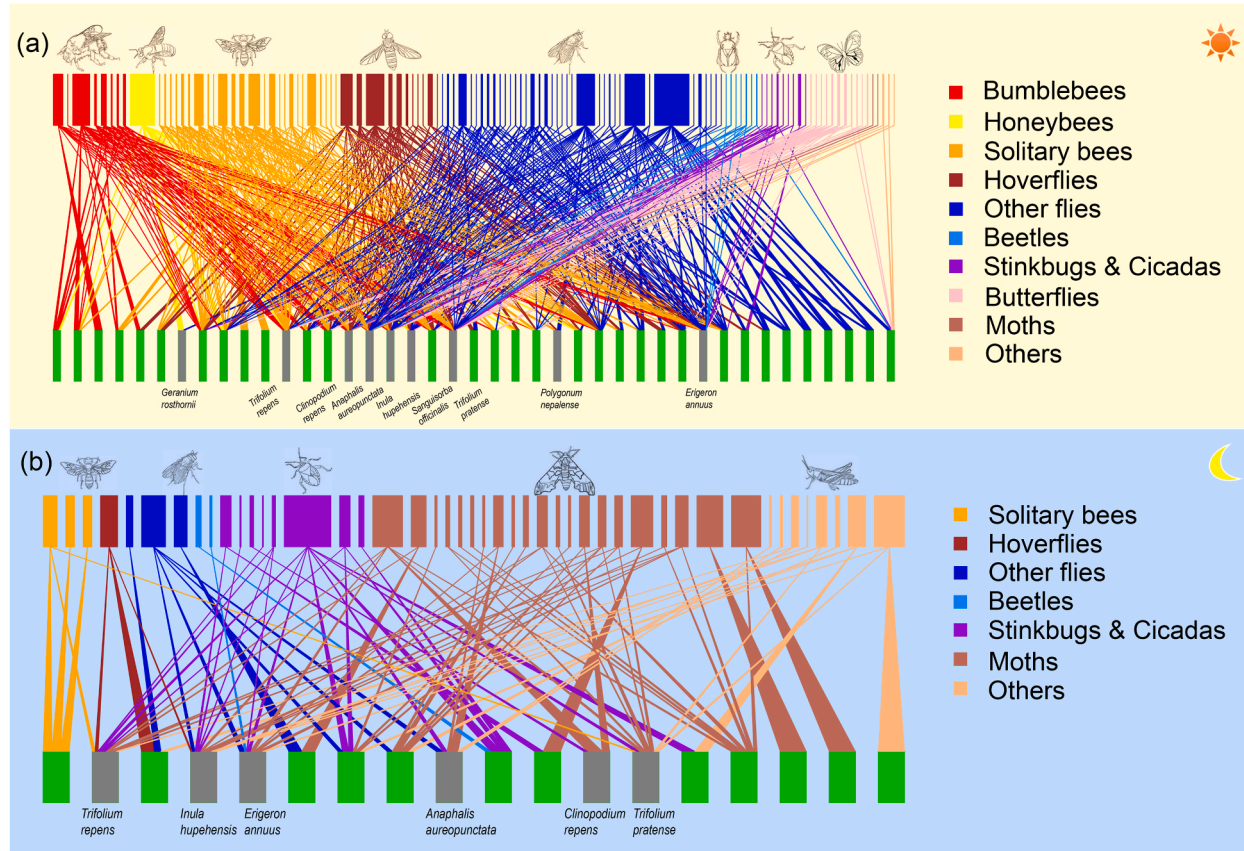


Fig. 1. Bipartite networks illustrating (a) diurnal and (b) nocturnal plant-pollinator interactions. The rectangles represent insect species (top) and plant species (bottom), and the connecting lines represent interactions among species. The width of the lines reflects the proportion of visits that the plant received from each pollinator species, while the boxes represent the cumulative proportion of visits to each plant by all insect species. Species used for subsequent pollen deposition and germination analysis are indicated by grey rectangles.

Table 2
Species-level metrics for diurnal and nocturnal networks of nine dominant plant species. Species strength is the sum of dependencies of each species, and d' is species-level specialisation.

	Plant		Pollinator	
	Species strength	d'	Species strength*	$d'†$
Diurnal				
<i>Sanguisorba officinalis</i>	5.50	0.27	1.24 ± 0.33	0.44 ± 0.02
<i>Polygonum nepalense</i>	4.85	0.25	1.02 ± 0.23	0.42 ± 0.02
<i>Trifolium pratense</i>	9.09	0.21	0.67 ± 0.14	0.40 ± 0.02
<i>Trifolium repens</i>	7.84	0.28	0.72 ± 0.17	0.42 ± 0.03
<i>Clinopodium repens</i>	4.53	0.18	0.76 ± 0.16	0.44 ± 0.03
<i>Geranium rosthornii</i>	1.92	0.40	1.21 ± 0.30	0.43 ± 0.04
<i>Anaphalis aureopunctata</i>	13.26	0.24	0.65 ± 0.16	0.37 ± 0.03
<i>Erigeron annuus</i>	9.39	0.23	0.73 ± 0.16	0.41 ± 0.03
<i>Inula hupehensis</i>	5.26	0.15	0.79 ± 0.16	0.44 ± 0.02
Nocturnal				
<i>Trifolium pratense</i>	2.87	0.46	0.63 ± 0.17	0.52 ± 0.04
<i>Trifolium repens</i>	3.34	0.42	0.55 ± 0.14	0.51 ± 0.03
<i>Clinopodium repens</i>	2.24	0.83	0.56 ± 0.30	0.57 ± 0.02
<i>Anaphalis aureopunctata</i>	4.26	0.74	0.56 ± 0.28	0.52 ± 0.03
<i>Erigeron annuus</i>	2.60	0.46	0.66 ± 0.18	0.50 ± 0.03
<i>Inula hupehensis</i>	3.76	0.33	0.55 ± 0.12	0.55 ± 0.04

†The measure of species-level specialisation of all pollinators (mean±se) that visited the correlate plant species.

* The species strength of all pollinators (mean±se) that visited the correlate plant species.

network, with *G. rosthornii* showing the highest d' value of 0.40. Its main diurnal visitors were *Apis cerana* individuals. *Anaphalis aureopunctata* was the most important contributor to link weighting with the visitor community, with the highest species strength in both diurnal and nocturnal networks (Table 2), likely due to its specific dioecious sexual system.

3.2. Comparing plant reproduction effects of diurnal and nocturnal pollinators

The number of pollen grains deposited during the daytime was significantly higher than that deposited at night ($t = -3.0$, $df = 8$, $P = 0.02$; Fig. 2a), and similar results were found for the number of pollen grains germinating ($t = -2.7$, $df = 8$, $P = 0.03$; Fig. 2b). However, the contribution of nocturnal pollinators to the reproduction of each plant might be species-specific (Fig. 3; Table A.10). Based on diurnal and nocturnal pollen deposition data, the nine species in our study can be divided into three groups (Fig. 3): the first group (*P. nepalense* and *S. officinalis* (Fig. 3a-b)) was referred to as autogamous plants (i.e., the pollen deposition and germination numbers were not significantly different between the autogamy and control groups), the second group (*Trifolium pratense*, *Trifolium repens*, *Clinopodium repens*, and *G. rosthornii* (Fig. 3c-f)) was referred to as diurnal plants (with pollen deposition and germination numbers that were significantly higher during the day than at night), and the third group (*I. hupehensis*, *E. annuus*, and *A. aureopunctata* (Fig. 3g-i)) was referred to as 24-h plants (i.e., the pollen deposition and germination numbers had no significant difference between the day and night).

We compared the germination rates of three 24-h plants (*I. hupehensis*, *E. annuus*, and *A. aureopunctata*) which rely to a large extent on both diurnal and nocturnal pollinators for their reproduction. We found that the nocturnal pollen germination rate of *I. hupehensis* was significantly higher than that during the day ($t = 2.2$, $df = 87$, $P = 0.03$; Fig. 3l). Although there was no significant difference in

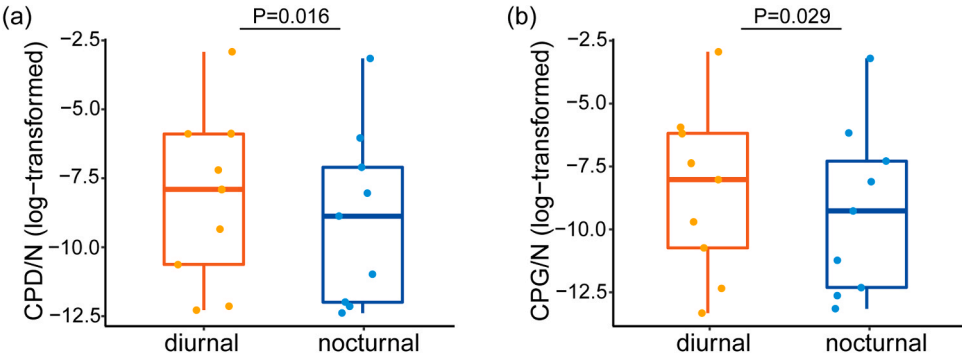
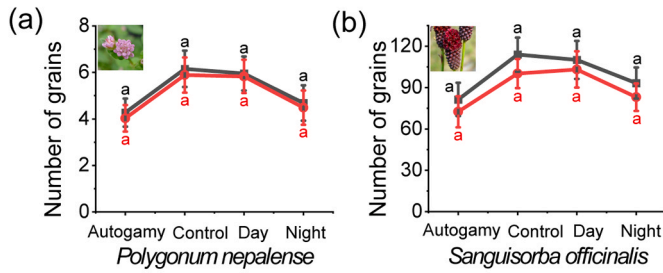
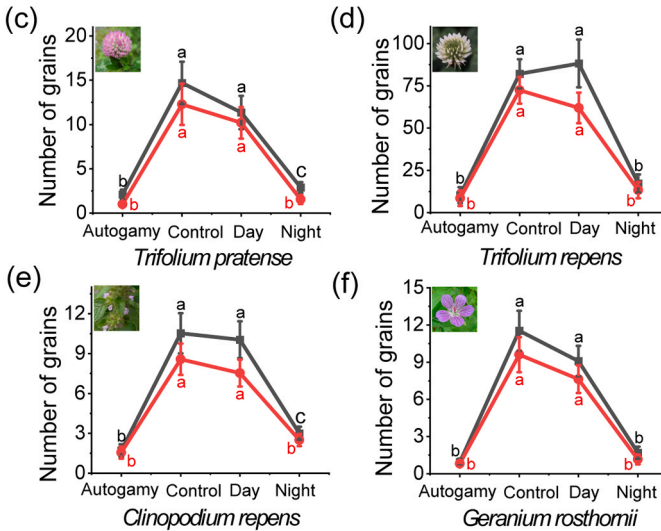


Fig. 2. The conspecific pollen (a) deposition (CPD) and (b) germination (CPG) (divided by the number of pollen grains per flower (N) for stand-ardisation) between the diurnal and nocturnal periods in the Dajihu wetland. The middle lines of the boxplots represent the mean for each variable in the station while the lines delimit the first (bottom line) and the third (top line) quartiles. Data from calculations of nine dominant plants are represented by dots and log10-transformed.

Autonomous Plants



Diurnal Plants



24-h Plants

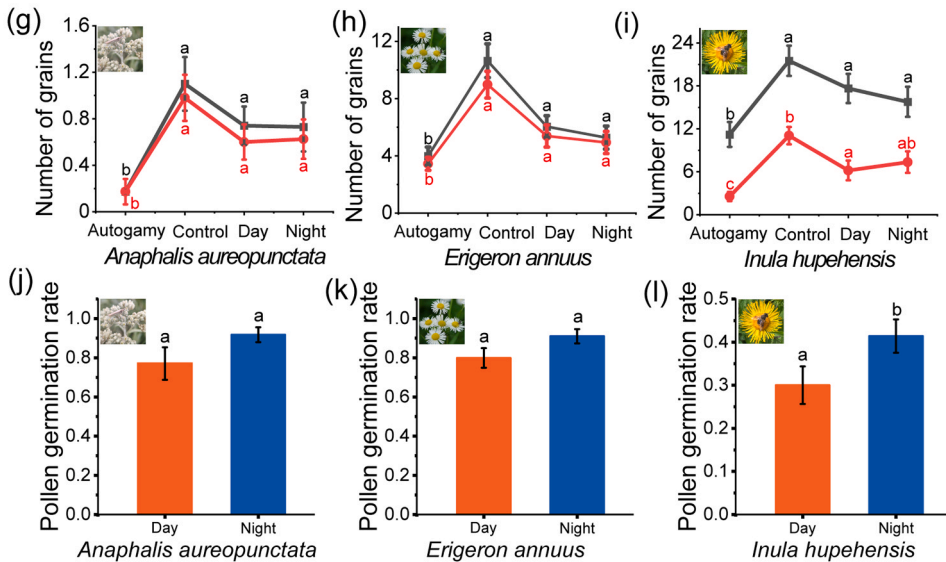


Fig. 3. Mean (+SE) number of pollen grains deposited and germinating among the pollination treatments for nine dominant plant species (a-i). The black diamonds represent the number of pollen grains deposited, and the red circles represent the number of pollen grains germinating on stigmas. Different letters in black or red respectively indicate significant differences using Tukey's test after GLMM. Mean (+SE) diurnal and nocturnal germination rate of pollen grains deposited on stigmas of 24-h plants (which rely on both nocturnal and diurnal pollinators for reproduction) (j-l). Different letters indicate significant differences.

the nocturnal and diurnal pollen germination rates of *E. annuus* ($t = 1.6$, $df = 84$, $P = 0.12$; Fig. 3k) and *A. aureopunctata* ($t = 1.4$, $df = 31$, $P = 0.16$; Fig. 3j), their nocturnal pollen germination rates were still higher than their diurnal pollen germination rates (Table A.11).

4. Discussion

Our abilities to extract network data from collected nocturnal pollinators revealed their substantial contribution to the ecological integrity of communities. Few pollination surveys have been conducted at the community level in Shennongjia. Some pollinators and plants recorded in the diurnal network in this study overlapped with data recorded in the Shennongjia area, but because only the diurnal pollinator network was studied, the many nocturnal pollinators were not included (Du et al., 2007). In this study, we recorded more pollinator groups, sampling a total of 55 families of visitors, comprising 44 families of diurnal pollinators and 28 families of nocturnal pollinators. Likewise, the recorded plant species are also in general agreement with those surveyed by Luo et al. (2015) for the wetland flora of Dajiuahu (except for plant species not visited by pollinators). Although moths are usually neglected from pollination studies, many moth species are generalist nectarivores as adults, and they may serve as important secondary pollinators for diurnally pollinated plants (Macgregor and Scott-Brown, 2020). This means that they may help to ensure the reproductive success of these plants, by providing an alternative means of pollination. In addition, the response to environmental changes varies among different taxa within the interaction networks, reflecting their diverse sensitivities (Pocock et al., 2012). This suggests that the extent to which pollination interactions between day and night are affected by environmental changes may differ. Considering the current rates of species decline and extinction and its impact on ecosystem function (Vamوسي et al., 2006), a more comprehensive understanding of the pollination network within the entire network can help to improve our understanding of the critical contributors to ecosystem services (Vamوسي et al., 2017).

At the network level, the diurnal and nocturnal pollinator networks were highly specialised. However, at the species level, the diurnal pollinators appeared to be highly generalised compared to the nocturnal pollinators. Differences in pollination syndromes between diurnal and nocturnal pollinators could be influencing the degree of specialisation in the nocturnal pollinators. The higher the specialisation level, the smaller the foraging width of the visitors (Fontaine et al., 2008). In this sense, the pollination breadth of diurnal bees and flies tend to be more extensive than those of nocturnal moths in terms of colour, odour type, shape, orientation, size, symmetry, and reward (Rosas-Guerrero et al., 2014). This highlights the potential vulnerability of nocturnal pollinators to the temporary or permanent lack of floral resources in the area. Meanwhile, both diurnal and nocturnal networks were significantly nested, but the nestedness of the diurnal network was much higher than that of the nocturnal network. Nestedness increases with the complexity (i.e., number of interactions) of the network: for a given number of species, communities with more interactions are significantly more nested (Bascompte et al., 2003). Nestedness has also been shown to increase network robustness, as nested networks appear less prone to the detrimental effects of habitat loss and species extinctions (Fortuna et al., 2013; Kaiser-Bunbury et al., 2010). Lower nestedness suggests that the nocturnal pollination network might be less stable than the diurnal pollination network (Bascompte et al., 2003; Nielsen and Bascompte, 2007). These results suggest that the nocturnal network is more vulnerable and that the protection of nocturnal pollinators as well as plant species where they forage should be strengthened. Meanwhile, the overlapping diurnal and nocturnal networks will mutually reinforce each other (Giavi et al., 2021; Knop et al., 2017), so better protection of nocturnal pollinators would also be beneficial for diurnal pollinators and the reverse is also true.

By comparing species-level parameters, we also found that specialisation of the pollinator and plant is determined by the period (day or night). However, some studies on the pollination efficiency of diurnal and nocturnal pollinators of a single plant species have found that, although there are more frequent visits by pollinators during the day, pollinators at night are more efficient (Dar et al., 2006; Jaca et al., 2020; Jennersten and Morse, 1991; Miyake and Yahara, 1999; Young, 2002). In this study, the species strengths of the nine dominant plant species and their visitors in the diurnal network were relatively higher than those in the nocturnal network (Table 2). However, a comparison of diurnal and nocturnal pollen deposition data for the nine plant species showed that not all species had a greater quantity of pollen deposited during the day than at night (Fig. 3). This means that plant-pollinator interaction network data may not be sufficient to predict the relative contribution of diurnal and nocturnal pollinators to the reproduction of each plant species. We suggest that additional informative variables should be recorded, such as seed set counts, as a supplementary measure of the effectiveness of pollination deposition events (Alison et al., 2022). Simultaneously, factors such as flower abundance, flower shape, and even phylogenetic relationships can influence the species-level parameters of a species in both diurnal and nocturnal networks (Vamوسي et al., 2014). However, considering all these multifaceted factors would make this a more complex issue, and we hope for more in-depth research in the future.

Based on diurnal plant-pollinator interactions data, we found that the major pollinator group of diurnal plants (*T. pretence*, *T. repens*, *C. repens* and *G. rosthornii*) was comprised of Hymenoptera (bumblebees, honeybees, solitary bees), whereas flies formed the major pollinator group of the 24-h plants (*I. hupehensis*, *E. annuus*, and *A. aureopunctata*) (Fig. A.5). Many researchers have stated that bees are more likely than flies to receive pollen on their bodies and thus to move conspecific pollen between flowers (Ballantyne et al., 2015; Garratt et al., 2014; Rader et al., 2016, 2013, 2009). Hence, we expected that nocturnal pollinators would contribute more to plant reproduction where diurnal pollinators were not effective. In addition, in the autonomous plants, *Polygonum nepalense* exhibited autonomous self-pollination and *Sanguisorba officinalis* was predominately wind pollinated (Lloyd, 1979); insect pollination was only supplementary for these species. Reproduction events for such species are reportedly not affected by changes in pollinator groups but by abiotic factors or the physical distance between stigmas and anthers (Barrett, 2002; Dowding, 1987; Lloyd, 1979).

The extinction of the strongest interactors might induce declines in the robustness of pollination networks (Kaiser-Bunbury et al., 2010; Memmott et al., 2004). For species such as the 24-h plants in the third group, both nocturnal and diurnal pollinators are the key

taxa for ensuring their reproduction (Fig. 3); we suspect that the loss of either diurnal pollinators or nocturnal pollinators is likely to lead to the decline or even local extinction of the plant species in this community. Because of the rapid decline in pollinators, particularly wild bees (Biesmeijer et al., 2006; Clough et al., 2014; Potts et al., 2010), plants may receive pollination service from available fauna or abiotic factors to replace bees (Kaiser-Bunbury et al., 2010). Therefore, nocturnal pollinators may become increasingly important for plant reproduction. Focusing only on diurnal pollination may amplify the contribution of some pollinators to plant reproduction. In reality, nocturnal pollinators possess numerous advantages in facilitating plant reproduction. For example, direct competition for resources among most diurnal pollinators is avoided (Hegland et al., 2009), and the narrower foraging width of nocturnal pollinators can improve pollination efficiency and/or reduce the risk of heterospecific pollen interference (Brosi, 2016). Furthermore, global warming may increase the frequency and severity of extreme weather events, which can have a detrimental effect on plant reproduction (Mesihovic et al., 2016). High temperatures can adversely affect pollen development and germination (Mesihovic et al., 2016; Pressman, 2002). The lower temperatures during the night compared to the daytime can create more favourable conditions for pollen germination and may result in an increase in pollen germination rates and more fruit/seed production from a reproductive perspective. Therefore, nocturnal pollinators may become increasingly important for plant reproduction (Chamorro et al., 2012; Ramirez, 2004). Understanding both diurnal and nocturnal interactions between plants and pollinators in the community will help us provide more comprehensive and feasible management strategies, which is critical to ensuring the conservation of pollination services in the face of a series of threats brought about by environmental changes.

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CRediT authorship contribution statement

Vamosi Jana C.: Writing – review & editing, Validation, Supervision. **Wang Xiao-Fan:** Validation, Supervision, Resources, Project administration, Conceptualization. **Villalobos Soraya:** Writing – review & editing, Validation, Supervision, Data curation. **Gong Yan-Bing:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Conceptualization. **Teng Yue:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors have no conflicts of interest to declare.

Data availability

Data and R scripts are available on Figshare: 10.6084/m9.figshare.23004563.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.gecco.2024.e02798](https://doi.org/10.1016/j.gecco.2024.e02798).

References

- Alison, J., Alexander, J.M., Diaz Zeugin, N., Dupont, Y.L., Iseli, E., Mann, H.M.R., Høye, T.T., 2022. Moths complement bumblebee pollination of red clover: a case for day-and-night insect surveillance. *Biol. Lett.* 18, 20220187 <https://doi.org/10.1098/rsbl.2022.0187>.
- Ballantyne, G., Baldock, K.C.R., Willmer, P.G., 2015. Constructing more informative plant–pollinator networks: visitation and pollen deposition networks in a heathland plant community. *Proc. R. Soc. B Biol. Sci.* 282, 20151130. <https://doi.org/10.1098/rspb.2015.1130>.
- Barrett, S.C.H., 2002. The evolution of plant sexual diversity. *Nat. Rev. Genet.* 3, 274–284. <https://doi.org/10.1038/nrg776>.
- Bascompte, J., Jordano, P., Melian, C.J., Olesen, J.M., 2003. The nested assembly of plant–animal mutualistic networks. *Proc. Natl. Acad. Sci.* 100, 9383–9387. <https://doi.org/10.1073/pnas.1633576100>.
- Bates, D., Mächler, M., Bolker, B., Walker, S., 2015. Fitting linear mixed-effects models using lme4. *J. Stat. Softw.* 67 <https://doi.org/10.18637/jss.v067.i01>.

- Beckett, S.J., Boulton, C.A., Williams, H.T.P., 2014. FALCON: a software package for analysis of nestedness in bipartite networks. *F1000Research* 3, 185. <https://doi.org/10.12688/f1000research.4831.1>.
- Bersier, L.-F., Banašek-Richter, C., Cattin, M.-F., 2002. Quantitative descriptors of food-web matrices. *Ecology* 83, 2394–2407. [https://doi.org/10.1890/0012-9658\(2002\)083\[2394:QDOFWM\]2.0.CO;2](https://doi.org/10.1890/0012-9658(2002)083[2394:QDOFWM]2.0.CO;2).
- Biesmeijer, J.C., Roberts, S.P.M., Reemer, M., Ohlemüller, R., Edwards, M., Peeters, T., Schaffers, A.P., Potts, S.G., Kleukers, R., Thomas, C.D., Settele, J., Kunin, W.E., 2006. Parallel declines in pollinators and insect-pollinated plants in Britain and the Netherlands. *Science* 313, 351–354. <https://doi.org/10.1126/science.1127863>.
- Bosch, J., Martín González, A.M., Rodrigo, A., Navarro, D., 2009. Plant-pollinator networks: adding the pollinator's perspective. *Ecol. Lett.* 12, 409–419. <https://doi.org/10.1111/j.1461-0248.2009.01296.x>.
- Briolat, E.S., Gaston, K.J., Bennie, J., Rosenfeld, E.J., Troschianko, J., 2021. Artificial nighttime lighting impacts visual ecology links between flowers, pollinators and predators. *Nat. Community* 12, 4163. <https://doi.org/10.1038/s41467-021-24394-0>.
- Brosi, B.J., 2016. Pollinator specialization: from the individual to the community. *N. Phytol.* 210, 1190–1194. <https://doi.org/10.1111/nph.13951>.
- Byers, D.L., 2017. Studying plant–pollinator interactions in a changing climate: a review of approaches. *Appl. Plant Sci.* 5, 1700012. <https://doi.org/10.3732/app.1700012>.
- Chamorro, S., Heleno, R., Olesen, J.M., McMullen, C.K., Traveset, A., 2012. Pollination patterns and plant breeding systems in the Galápagos: a review. *Ann. Bot.* 110, 1489–1501. <https://doi.org/10.1093/aob/mcs132>.
- Cirtwill, A.R., Kaartinen, R., Rasmussen, C., Redr, D., Wirta, H., Olesen, J.M., Tiusanen, M., Ballantyne, G., Cunlold, H., Stone, G.N., Schmidt, N.M., Roslin, T., 2022. Stable pollination service in a generalist High Arctic community despite the warming climate. *Ecol. Monogr.* <https://doi.org/10.1002/ecm.1551>.
- Clough, Y., Ekroos, J., Báldi, A., Batáry, P., Bommarco, R., Gross, N., Holzschuh, A., Hopfenmüller, S., Knop, E., Kuussaari, M., Lindborg, R., Marini, L., Öckinger, E., Potts, S.G., Pöry, J., Roberts, S.P., Steffan-Dewenter, I., Smith, H.G., 2014. Density of insect-pollinated grassland plants decreases with increasing surrounding land-use intensity. *Ecol. Lett.* 17, 1168–1177. <https://doi.org/10.1111/ele.12325>.
- Corbet, S.A., Chen, F.-F., Chang, F.-F., Huang, S.-Q., 2020. Transient dehydration of pollen carried by hot bees impedes fertilization. *Arthropod-Plant Interact.* 14, 207–214. <https://doi.org/10.1007/s11829-019-09726-8>.
- Dar, S., Arizmendi, Ma. del C., Valiente-Banuet, A., 2006. Diurnal and nocturnal pollination of *marginatocereus marginatus* (Pachycereae: Cactaceae) in Central Mexico. *Ann. Bot.* 97, 423–427. <https://doi.org/10.1093/aob/mcj045>.
- Devoto, M., Bailey, S., Memmott, J., 2011. The ‘night shift’: nocturnal pollen-transport networks in a boreal pine forest. *Ecol. Entomol.* 36, 25–35. <https://doi.org/10.1111/j.1365-2311.2010.01247.x>.
- Dormann, C.F., 2011. How to be a specialist? Quantifying specialisation in pollination networks. *Netw. Biol.* 1, 1–20.
- Dormann, C.F., Strauss, R., 2014. A method for detecting modules in quantitative bipartite networks. *Methods Ecol. Evol.* 5, 90–98. <https://doi.org/10.1111/2041-210X.12139>.
- Dormann, C.F., Gruber, B., Fründ, J., 2008. Introducing the bipartite package: analysing ecological networks. *R. N.* 8, 8–11.
- Dowding, P., 1987. Wind pollination mechanisms and aerobiology. *Int. Rev. Cytology* 107, 421–437. [https://doi.org/10.1016/S0074-7696\(08\)61084-0](https://doi.org/10.1016/S0074-7696(08)61084-0).
- Du, W., Wang, H.X., Wang, X.F., 2007. Insect visitors and their behaviors in the typical herbaceous plant communities of the Shennongjia Mountains. *Biodivers. Sci.* 15, 666. <https://doi.org/10.1360/biodiv.070177>.
- Fijen, T.P.M., Roovers, A., van Deijk, J., van Grunsven, R.H.A., 2023. Nocturnal pollination is equally important as, and complementary to, diurnal pollination for strawberry fruit production. *Agric. Ecosyst. Environ.* 350, 108475. <https://doi.org/10.1016/j.agee.2023.108475>.
- Fontaine, C., Collin, C.L., Dajoz, I., 2008. Generalist foraging of pollinators: diet expansion at high density: Flexibility of pollinator generalization. *J. Ecol.* 96, 1002–1010. <https://doi.org/10.1111/j.1365-2745.2008.01405.x>.
- Fortuna, M.A., Krishna, A., Bascompte, J., 2013. Habitat loss and the disassembly of mutualistic networks. *Oikos* 122, 938–942. <https://doi.org/10.1111/j.1600-0706.2012.00042.x>.
- Gaston, K.J., Bennie, J., 2014. Demographic effects of artificial nighttime lighting on animal populations. *Environ. Rev.* 22, 323–330. <https://doi.org/10.1139/er-2014-0005>.
- Giavi, S., Fontaine, C., Knop, E., 2021. Impact of artificial light at night on diurnal plant-pollinator interactions. *Nat. Community* 12, 1690. <https://doi.org/10.1038/s41467-021-22011-8>.
- Gong, Y.-B., Huang, S.-Q., 2009. Floral symmetry: pollinator-mediated stabilizing selection on flower size in bilateral species. *Proc. R. Soc. Lond. Ser. B Biol. Sci.* 276, 4013–4020. <https://doi.org/10.1098/rspb.2009.1254>.
- Gong, Y.-B., Huang, S.-Q., 2014. Interspecific variation in pollen-ovule ratio is negatively correlated with pollen transfer efficiency in a natural community. *Plant Biol.* 16, 843–847. <https://doi.org/10.1111/plb.12151>.
- Gong, Y.-B., Yang, M., Vamosi, J.C., Yang, H.-M., Mu, W.-X., Li, J.-K., Wan, T., 2016. Wind or insect pollination? Ambophily in a subtropical gymnosperm *Gnetum parvifolium* (Gnetales): ambophily in *Gnetum*. *Plant Species Biol.* 31, 272–279. <https://doi.org/10.1111/1442-1984.12112>.
- Hegland, S.J., Grytnes, J.-A., Totland, Ø., 2009. The relative importance of positive and negative interactions for pollinator attraction in a plant community. *Ecol. Res.* 24, 929–936. <https://doi.org/10.1007/s11284-008-0572-3>.
- Jaca, J., Nogales, M., Traveset, A., 2020. Effect of diurnal vs. nocturnal pollinators and flower position on the reproductive success of *Echium simplex*. *Arthropod-Plant Interact.* 14, 409–419. <https://doi.org/10.1007/s11829-020-09759-4>.
- Jennersten, O., Morse, D.H., 1991. The quality of pollination by diurnal and nocturnal insects visiting common milkweed, *Asclepias syriaca*. *Am. Midl. Nat.* 125, 18. <https://doi.org/10.2307/2426365>.
- Kaiser-Bunbury, C.N., Muff, S., Memmott, J., Müller, C.B., Caflisch, A., 2010. The robustness of pollination networks to the loss of species and interactions: a quantitative approach incorporating pollinator behaviour. *Ecol. Lett.* 13, 442–452. <https://doi.org/10.1111/j.1461-0248.2009.01437.x>.
- Kearns, C.A., Inouye, D.W., 1993. *Techniques for Pollination Biologists*. University Press of Colorado.
- Knop, E., Zoller, L., Ryser, R., Gerpe, C., Hörler, M., Fontaine, C., 2017. Artificial light at night as a new threat to pollination. *Nature* 548, 206–209. <https://doi.org/10.1038/nature23288>.
- Knop, E., Gerpe, C., Ryser, R., Hofmann, F., Menz, M.H.M., Trösch, S., Ursenbacher, S., Zoller, L., Fontaine, C., 2018. Rush hours in flower visitors over a day-night cycle. *Insect Conserv. Divers.* 11, 267–275. <https://doi.org/10.1111/icad.12277>.
- Lloyd, D.G., 1979. Some reproductive factors affecting the selection of self-fertilization in plants. *Am. Nat.* 113, 67–79. <https://doi.org/10.1086/283365>.
- Luo, T., Lun, Z.J., Gu, Y.S., Qin, Y.M., Zang, Z., Zhang, B., 2015. Plant community survey and ecological protection of Dajiuhu wetlands in Shennongjia area. *Wetl. Sci.* 13, 153–160.
- Macgregor, C.J., Scott-Brown, A.S., 2020. Nocturnal pollination: an overlooked ecosystem service vulnerable to environmental change. *Emerg. Top. Life Sci.* 4, 19–32. <https://doi.org/10.1042/ETLS20190134>.
- Memmott, J., Waser, N.M., Price, M.V., 2004. Tolerance of pollination networks to species extinctions. *Proc. R. Soc. B Biol. Sci.* 271, 2605–2611. <https://doi.org/10.1098/rspb.2004.2909>.
- Mesihovic, A., Iannaccone, R., Firon, N., Fragkostefanakis, S., 2016. Heat stress regimes for the investigation of pollen thermotolerance in crop plants. *Plant Reprod.* 29, 93–105. <https://doi.org/10.1007/s00497-016-0281-y>.
- Miyake, T., Yahara, T., 1999. Theoretical evaluation of pollen transfer by nocturnal and diurnal pollinators: when should a flower open? *Oikos* 86, 233. <https://doi.org/10.2307/3546441>.
- Nielsen, A., Bascompte, J., 2007. Ecological networks, nestedness and sampling effort. *J. Ecol.* 95, 1134–1141. <https://doi.org/10.1111/j.1365-2745.2007.01271.x>.
- Orford, K.A., Vaughan, I.P., Memmott, J., 2015. The forgotten flies: the importance of non-syrphid Diptera as pollinators. *Proc. R. Soc. B Biol. Sci.* 282, 20142934. <https://doi.org/10.1098/rspb.2014.2934>.
- Pocock, M.J.O., Evans, D.M., Memmott, J., 2012. The robustness and restoration of a network of ecological networks. *Science* 335, 973–977. <https://doi.org/10.1126/science.1214915>.

- Potts, S.G., Biesmeijer, J.C., Kremen, C., Neumann, P., Schweiger, O., Kunin, W.E., 2010. Global pollinator declines: trends, impacts and drivers. *Trends Ecol. Evol.* 25, 345–353. <https://doi.org/10.1016/j.tree.2010.01.007>.
- Potts, S.G., Imperatriz-Fonseca, V., Ngo, H.T., Aizen, M.A., Biesmeijer, J.C., Breeze, T.D., Dicks, L.V., Garibaldi, L.A., Hill, R., Settele, J., Vanbergen, A.J., 2016. Safeguarding pollinators and their values to human well-being. *Nature* 540, 220–229. <https://doi.org/10.1038/nature20588>.
- Pressman, E., 2002. The effect of heat stress on tomato pollen characteristics is associated with changes in carbohydrate concentration in the developing anthers. *Ann. Bot.* 90, 631–636. <https://doi.org/10.1093/aob/mcf240>.
- Rader, R., Howlett, B.G., Cunningham, S.A., Westcott, D.A., Newstrom-Lloyd, L.E., Walker, M.K., Teulon, D.A.J., Edwards, W., 2009. Alternative pollinator taxa are equally efficient but not as effective as the honeybee in a mass flowering crop. *J. Appl. Ecol.* 46, 1080–1087. <https://doi.org/10.1111/j.1365-2664.2009.01700.x>.
- Rader, R., Edwards, W., Westcott, D.A., Cunningham, S.A., Howlett, B.G., 2013. Diurnal effectiveness of pollination by bees and flies in agricultural Brassica rapa: implications for ecosystem resilience. *Basic Appl. Ecol.* 14, 20–27. <https://doi.org/10.1016/j.baae.2012.10.011>.
- Rader, R., Bartomeus, I., Garibaldi, L.A., Garratt, M.P.D., Howlett, B.G., Winfree, R., Cunningham, S.A., Mayfield, M.M., Arthur, A.D., Andersson, G.K.S., Bommarco, R., Brittain, C., Carvalheiro, L.G., Chacoff, N.P., Entling, M.H., Foully, B., Freitas, B.M., Gemmill-Herren, B., Ghazoul, J., Griffin, S.R., Gross, C.L., Herberthsson, L., Herzog, F., Hipólito, J., Jaggard, S., Jauker, F., Klein, A.-M., Kleijn, D., Krishnan, S., Lemos, C.Q., Lindström, S.A.M., Mandelik, Y., Monteiro, V.M., Nelson, W., Nilsson, L., Pattenmore, D.E., de, O., Pereira, N., Pisanty, G., Potts, S.G., Reemer, M., Rundlöf, M., Sheffield, C.S., Scheper, J., Schüepp, C., Smith, H.G., Stanley, D.A., Stout, J.C., Szentgyörgyi, H., Taki, H., Vergara, C.H., Viana, B.F., Woyciechowski, M., 2016. Non-bee insects are important contributors to global crop pollination. *Proc. Natl. Acad. Sci.* 113, 146–151. <https://doi.org/10.1073/pnas.1517092112>.
- Ramirez, N., 2004. Pollination specialization and time of pollination on a tropical Venezuelan plain: variations in time and space. *Bot. J. Linn. Soc.* 145, 1–16. <https://doi.org/10.1111/j.1095-8339.2004.00181.x>.
- Rosas-Guerrero, V., Aguilar, R., Martén-Rodríguez, S., Ashworth, L., Lopezaraiza-Mikel, M., Bastida, J.M., Quesada, M., 2014. A quantitative review of pollination syndromes: do floral traits predict effective pollinators? *Ecol. Lett.* 17, 388–400. <https://doi.org/10.1111/ele.12224>.
- Vamosi, J.C., Knight, T.M., Steets, J.A., Mazer, S.J., Burd, M., Ashman, T.-L., 2006. Pollination decays in biodiversity hotspots. *Proc. Natl. Acad. Sci.* 103, 956–961. <https://doi.org/10.1073/pnas.0507165103>.
- Vamosi, J.C., Moray, C.M., Garcha, N.K., Chamberlain, S.A., Mooers, A.Ø., 2014. Pollinators visit related plant species across 29 plant–pollinator networks. *Ecol. Evol.* 4, 2303–2315. <https://doi.org/10.1002/ece3.1051>.
- Vamosi, J.C., Gong, Y., Adamowicz, S.J., Packer, L., 2017. Forecasting pollination declines through DNA barcoding: the potential contributions of macroecological and macroevolutionary scales of inquiry. *N. Phytol.* 214, 11–18. <https://doi.org/10.1111/nph.14356>.
- Vara Prasad, P., 1999. Fruit number in relation to pollen production and viability in groundnut exposed to short episodes of heat stress. *Ann. Bot.* 84, 381–386. <https://doi.org/10.1006/anbo.1999.0926>.
- Walton, R.E., Sayer, C.D., Bennion, H., Axmacher, J.C., 2020. Nocturnal pollinators strongly contribute to pollen transport of wild flowers in an agricultural landscape. *Biol. Lett.* 16, 20190877. <https://doi.org/10.1098/rsbl.2019.0877>.
- Wei, N., Kaczorowski, R.L., Arceo-Gómez, G., O'Neill, E.M., Hayes, R.A., Ashman, T.-L., 2021. Pollinators contribute to the maintenance of flowering plant diversity. *Nature* 597, 688–692. <https://doi.org/10.1038/s41586-021-03890-9>.
- Wood, T.J., Holland, J.M., Goulson, D., 2015. A comparison of techniques for assessing farmland bumblebee populations. *Oecologia* 177, 1093–1102. <https://doi.org/10.1007/s00442-015-3255-0>.
- Young, H.J., 2002. Diurnal and nocturnal pollination of *Silene alba* (Caryophyllaceae). *Am. J. Bot.* 89, 433–440. <https://doi.org/10.3732/ajb.89.3.433>.